

Recursion

Function calling itself

A() {

A() {

B() {

A() → recursive call

}

}

① Smaller input → ✓

② N = 1 → ✓

③ (N-1) !

PMI

$$\sum_{i=1}^N i \Rightarrow 1+2+3+\dots+N = \frac{N(N+1)}{2}$$

$$i=1 \Rightarrow i \Rightarrow 1 = \frac{1 \times (1+1)}{2} = \checkmark$$

$$i=1$$

$$\sum_{i=1}^2 i \Rightarrow 1+2 = \frac{2 \times (2+1)}{2} = \checkmark$$

$$i=1$$

$$\sum_{i=1}^{N+1} i \Rightarrow 1+2+3+\dots+(N+1) = \frac{(N+1)(N+2)}{2}$$

$$N+1$$

$$\sum_{i=1}^{N+1} i = 1+2+3+\dots+(N+1) = \frac{(N+1)(N+2)}{2}$$

$$i=1$$

$$\frac{N(N+1) + (N+1)}{2} =$$

$$(N+1)(N+1) = \frac{(N+1)(N+2)}{2}$$

Recursion + Recursion

① N! = (N-1)! * N → Recursion

② SP + self work → ✓

Box

$$5! = 4! \times 5$$

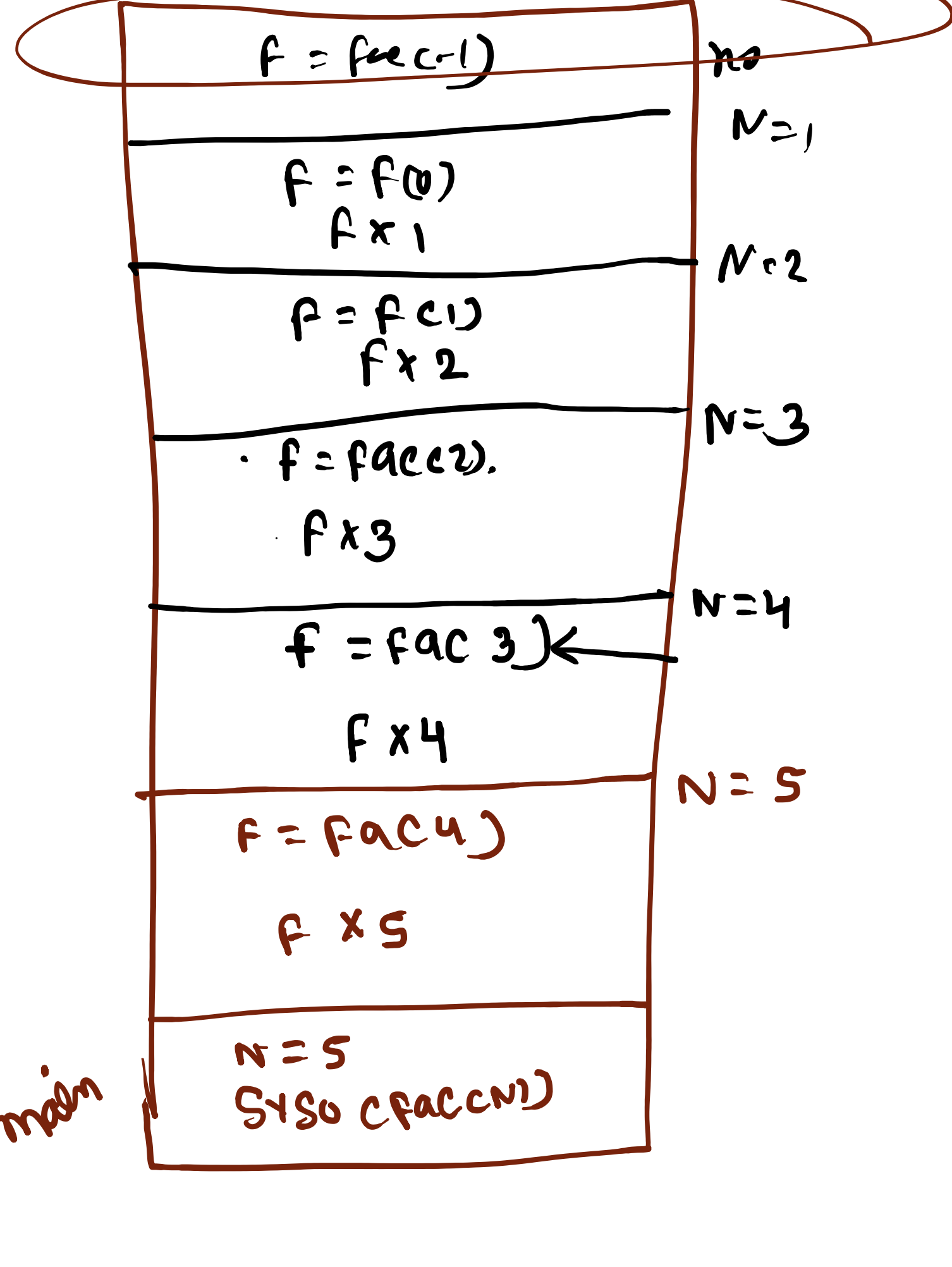
$$G \leftarrow 3! \times 4$$

$$G \leftarrow 2! \times 3$$

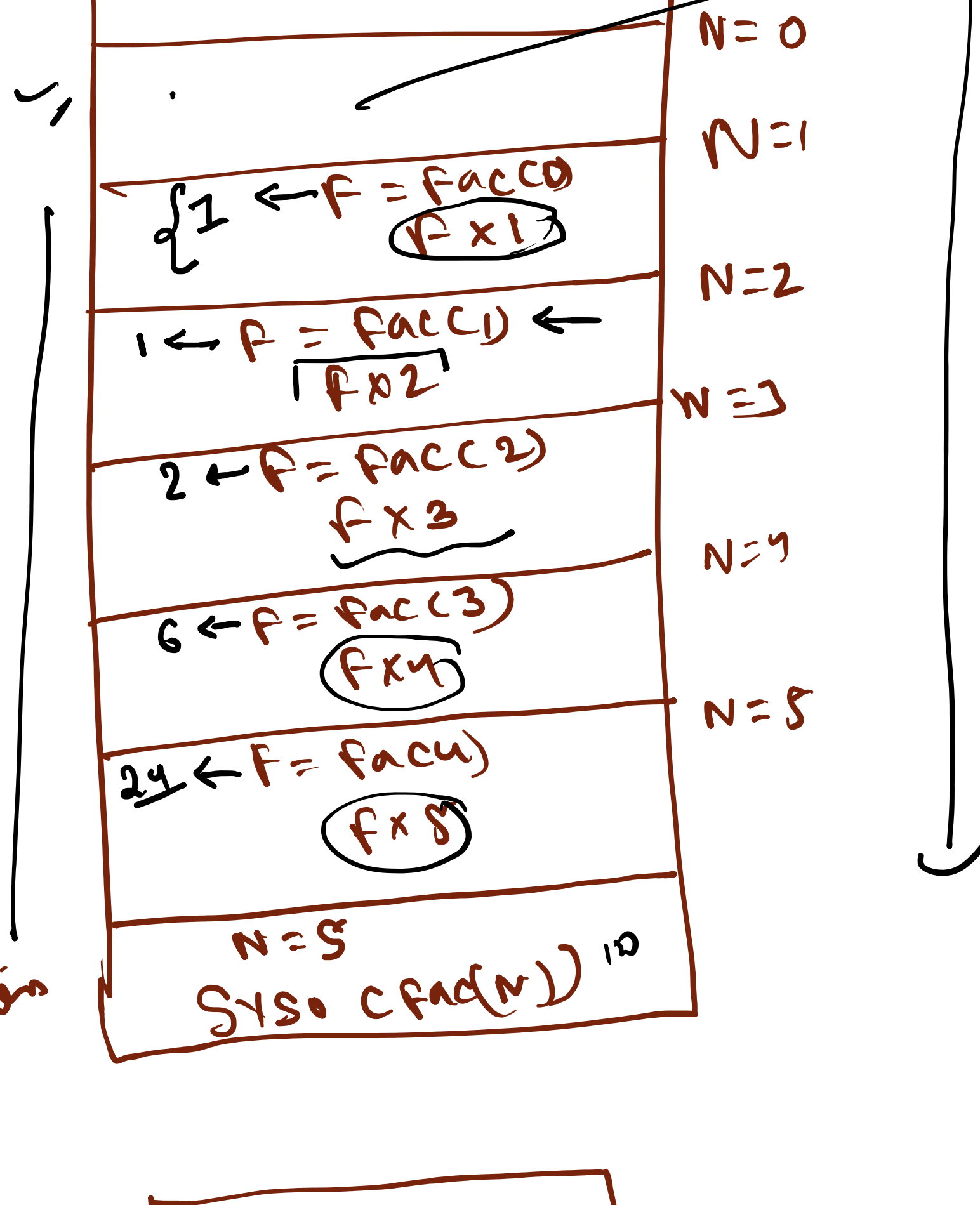
$$2 \leftarrow 1! \times 2$$

$$1 \leftarrow 0! \times 1$$

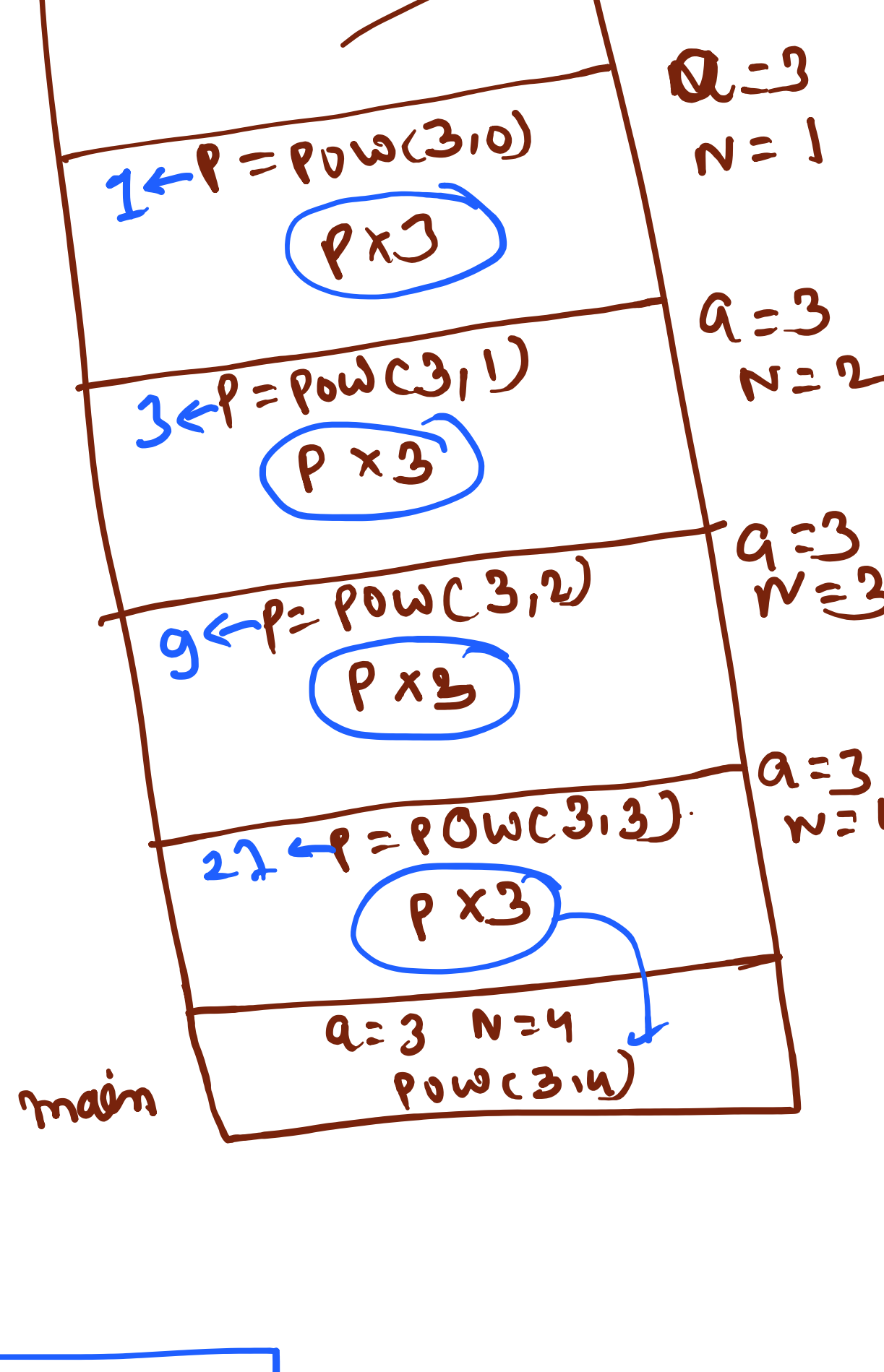
```
public static void main(String[] args) {  
    // TODO Auto-generated method stub  
    int n = 5;  
    System.out.println(fac(n));  
}  
  
public static int fac(int n) {  
    // TODO Auto-generated method stub  
    int f = fac(n-1); // SP  
    return f * n;  
}
```



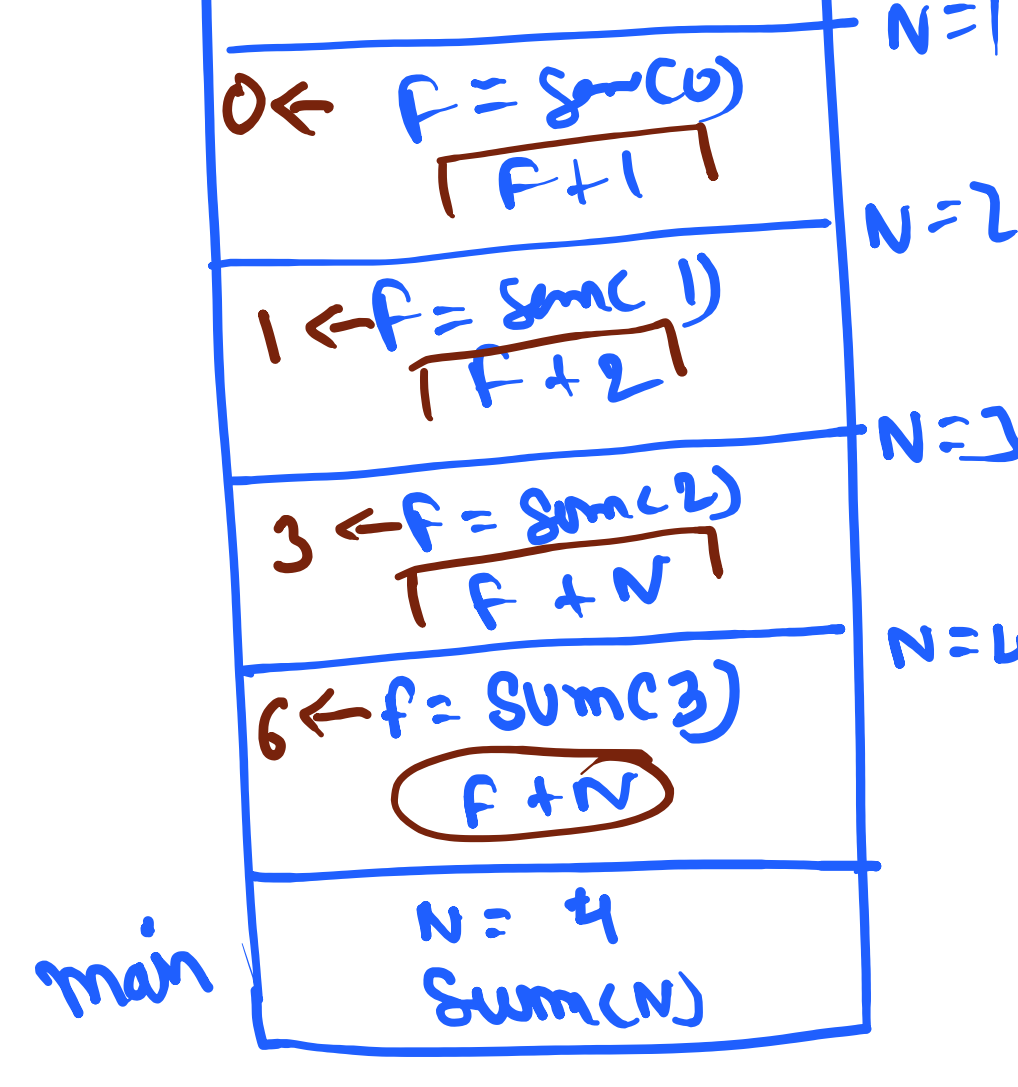
```
public static void main(String[] args) {  
    // TODO Auto-generated method stub  
    int n = 5;  
    System.out.println(fac(n));  
}  
  
public static int fac(int n) {  
    // TODO Auto-generated method stub  
    if (n == 0) {  
        return 1;  
    }  
    int f = fac(n-1); // SP  
    return f * n;  
}
```



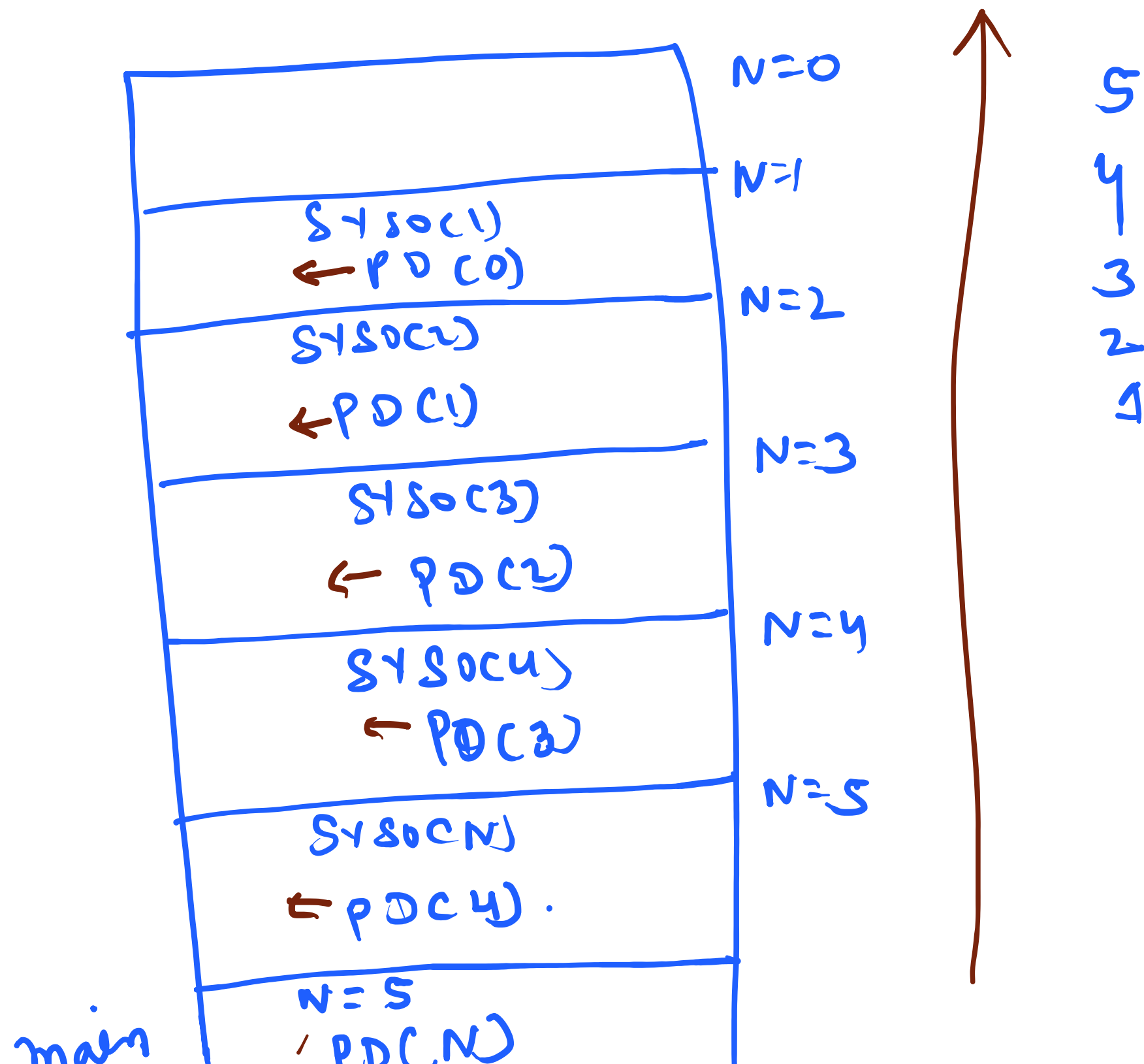
```
public static void main(String[] args) {  
    // TODO Auto-generated method stub  
    int a = 3;  
    int n = 4;  
    System.out.println(pow(a, n));  
}  
  
public static int pow(int a, int n) {  
    if (n == 0) {  
        return 1;  
    }  
    int p = pow(a, n-1);  
    return p * a;  
}
```



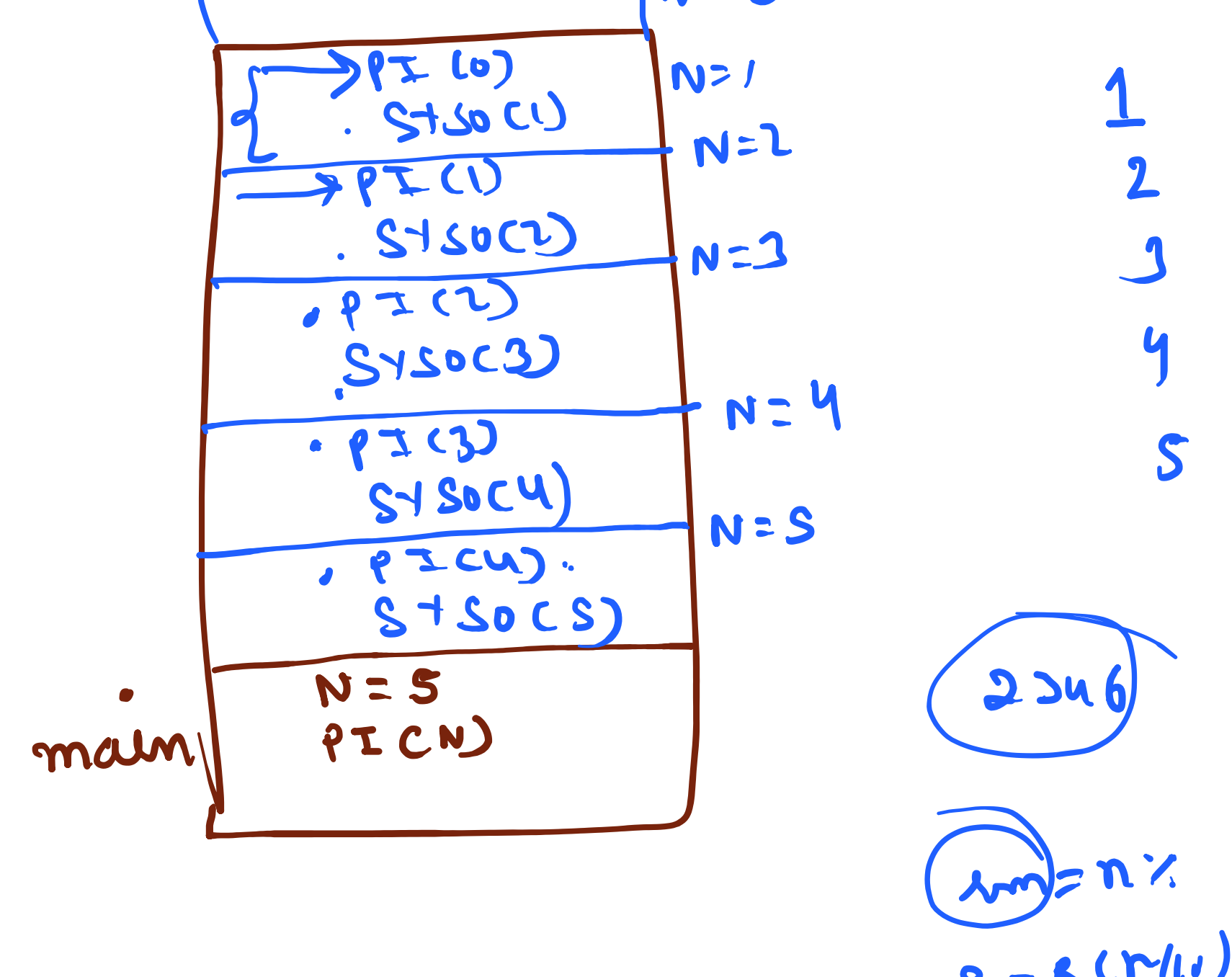
```
public static int Sum(int n) {  
    if (n == 0) {  
        return 0;  
    }  
    int f = Sum(n-1);  
    return f + n;  
}
```



```
public static void PD(int n) {  
    if (n == 0) {  
        return;  
    }  
    System.out.println(n);  
    PD(n-1);  
}
```

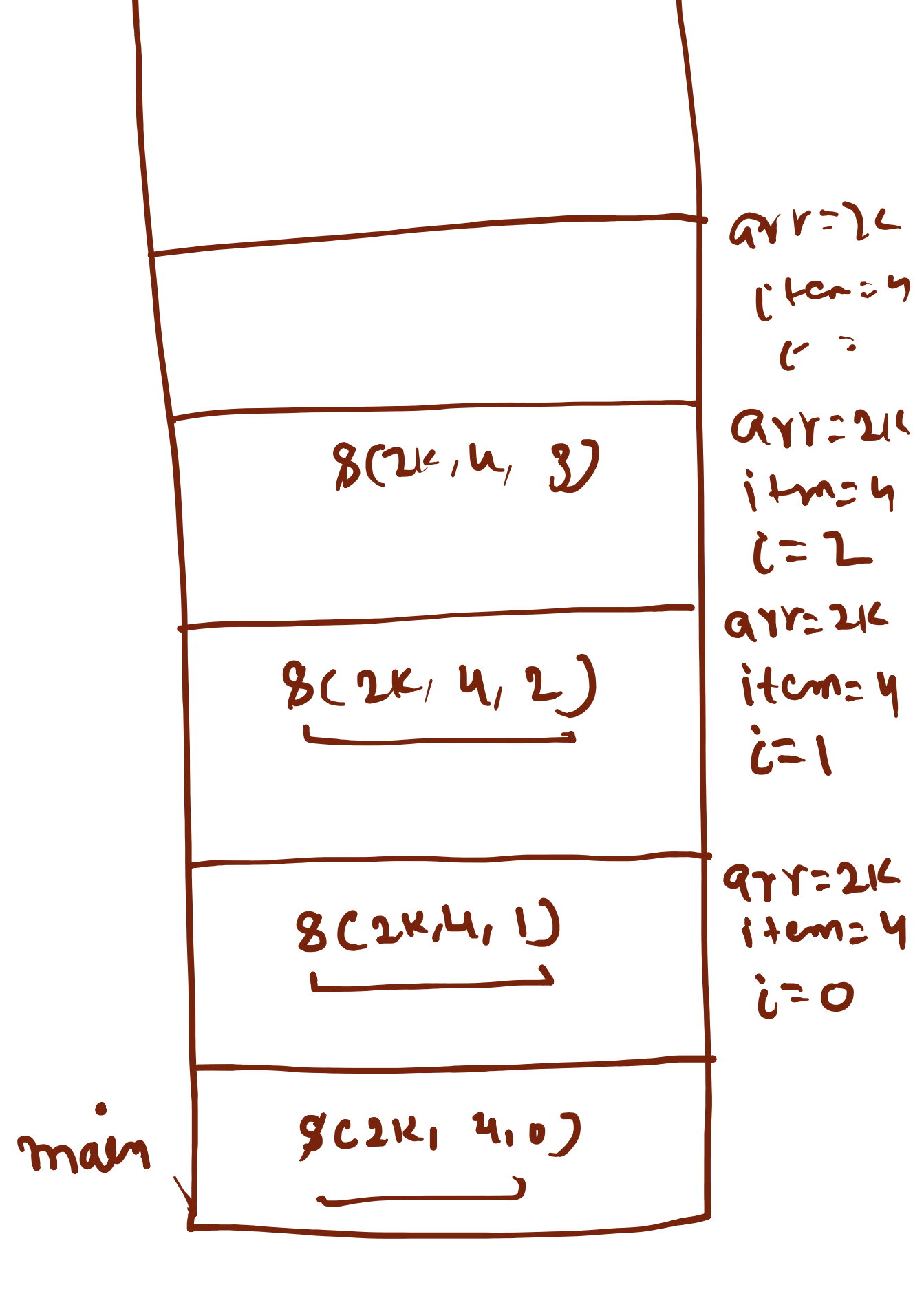


```
public static void main(String[] args) {  
    int n = 5;  
    PI(n);  
}  
  
public static void PI(int n) {  
    if (n == 0) {  
        return;  
    }  
    PI(n-1);  
    System.out.println(n);  
}
```

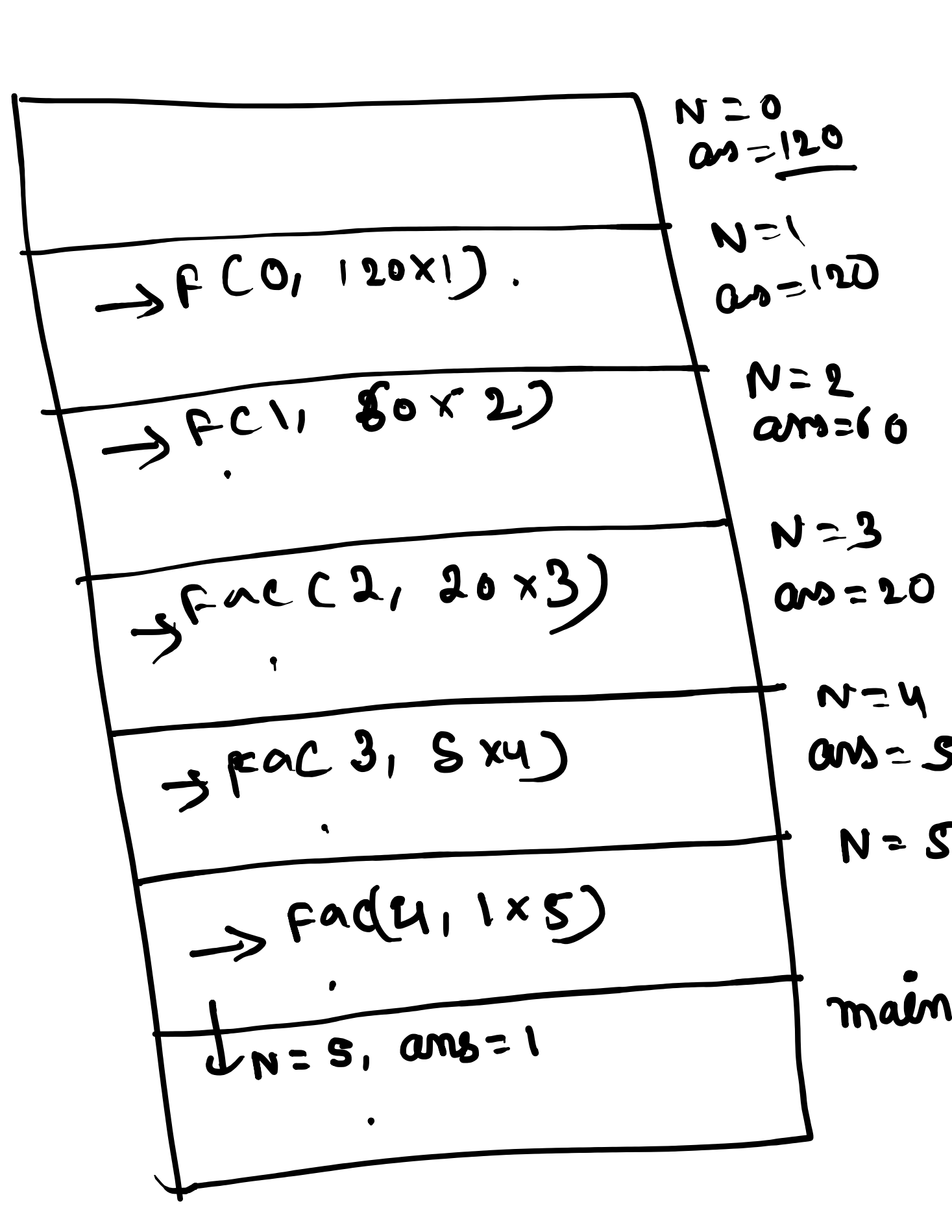


```
int[] arr = {2, 3, 4, 5, 4};
```

```
public static void main(String[] args) {  
    // TODO Auto-generated method stub  
    int[] arr = {2, 3, 4, 5, 4};  
    int item = 4;  
    System.out.println(search(arr, item, 0));  
}  
  
public static int search(int[] arr, int item, int i) {  
    if (arr[i] == item) {  
        return i;  
    }  
    return search(arr, item, i+1);  
}
```



```
public static int fac(int n, int ans) {  
    // TODO Auto-generated method stub  
    if (n == 0) {  
        return ans;  
    }  
    return fac(n-1, ans*n);  
}
```

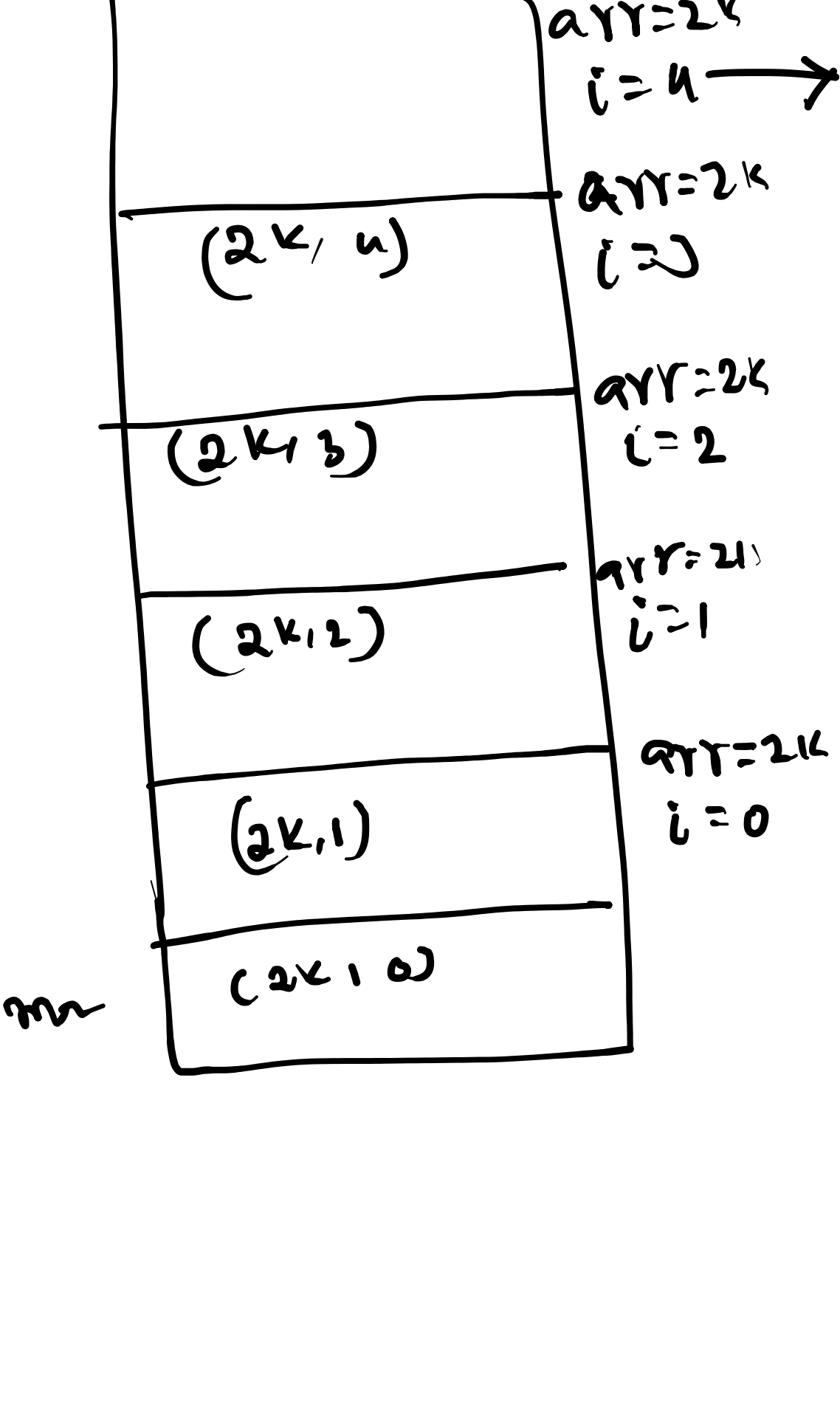


```
int[] arr = {3, 5, 7, 8, 0};
```

arr[0] > arr[1]

arr[1] > arr[2]

```
public static boolean Is_Sorted(int[] arr, int i) {  
    if (arr[i] > arr[i+1]) {  
        return false;  
    }  
    return Is_Sorted(arr, i+1);  
}
```



```
public static int countVowels(String s) {  
    char ch = s.charAt(0);  
    int c = 0;  
    if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u') {  
        c = 1;  
    }  
    return c + countVowels(s.substring(1));  
}
```

